Kinematic Analysis Report

Auto-Solver

February 3, 2026

1 Step 1: DH Matrices

Matrix T_1

$$\begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & L\cos(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) & 0 & L\sin(\theta_1) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_2

$$\begin{bmatrix} \cos(\theta_2) & 0 & -\sin(\theta_2) & 2L\cos(\theta_2) \\ \sin(\theta_2) & 0 & \cos(\theta_2) & 2L\sin(\theta_2) \\ 0 & -1 & 0 & 5L \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_3

$$\begin{bmatrix} -\cos(\theta_3) & 0 & -\sin(\theta_3) & 0 \\ -\sin(\theta_3) & 0 & \cos(\theta_3) & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_4

$$\begin{bmatrix} \cos(\theta_4) & 0 & -\sin(\theta_4) & \frac{5L\cos(\theta_4)}{2} \\ \sin(\theta_4) & 0 & \cos(\theta_4) & \frac{5L\sin(\theta_4)}{2} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_5

$$\begin{bmatrix} \cos(\theta_5) & 0 & \sin(\theta_5) & \frac{5L\cos(\theta_5)}{2} \\ \sin(\theta_5) & 0 & -\cos(\theta_5) & \frac{5L\sin(\theta_5)}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_6

$$\begin{bmatrix} -\cos(\theta_6) & 0 & \sin(\theta_6) & 0 \\ -\sin(\theta_6) & 0 & -\cos(\theta_6) & 0 \\ 0 & -1 & 0 & \frac{5L}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_7

$$\begin{bmatrix} \cos(\theta_7) & 0 & -\sin(\theta_7) & 2L\cos(\theta_7) \\ \sin(\theta_7) & 0 & \cos(\theta_7) & 2L\sin(\theta_7) \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2 Step 2: Loop Equation

$$\prod_{i=1}^{7} T_i = I$$

3 Step 3: Permutations

Set 1: $T_1T_2T_3T_4 = T_7^{-1}T_6^{-1}T_5^{-1}$

Set 2: $T_2T_3T_4T_5 = T_1^{-1}T_7^{-1}T_6^{-1}$

Set 3: $T_3T_4T_5T_6 = T_2^{-1}T_1^{-1}T_7^{-1}$

Set 4: $T_4T_5T_6T_7 = T_3^{-1}T_2^{-1}T_1^{-1}$

Set 5: $T_5T_6T_7T_1 = T_4^{-1}T_3^{-1}T_2^{-1}$

Set 6: $T_6T_7T_1T_2 = T_5^{-1}T_4^{-1}T_3^{-1}$

Set 7: $T_7T_1T_2T_3 = T_6^{-1}T_5^{-1}T_4^{-1}$

4 Step 4: Simplification

Equations simplified.

Set 1

$$\frac{L\left(-5\sin(\theta_4)\sin(\theta_1+\theta_2) - 5\sin(\theta_7) + 2\cos(\theta_1) - 5\cos(\theta_3)\cos(\theta_4)\cos(\theta_1+\theta_2) - 5\cos(\theta_6)\cos(\theta_7) + 4\cos\right.}{2}$$

$$\sin(\theta_1)\sin(\theta_3)\cos(\theta_2) + \sin(\theta_2)\sin(\theta_3)\cos(\theta_1) + \sin(\theta_5)\sin(\theta_6) = 0$$

$$-\sin(\theta_1)\sin(\theta_2)\cos(\theta_4) + \sin(\theta_1)\sin(\theta_4)\cos(\theta_2)\cos(\theta_3) + \sin(\theta_2)\sin(\theta_4)\cos(\theta_1)\cos(\theta_3) + \cos(\theta_1)\cos(\theta_2)\cos$$

$$\frac{L\left(2\sin(\theta_1) + 5\sin(\theta_4)\cos(\theta_1+\theta_2) - 5\sin(\theta_6) - 5\sin(\theta_1+\theta_2)\cos(\theta_3)\cos(\theta_4) + 4\sin(\theta_1+\theta_2)\right)}{2} = 0$$

$$\frac{\sin(\theta_5)\sin^2(\theta_6)\sin(\theta_7)}{\cos(\theta_6)} - \frac{\sin(\theta_5)\sin(\theta_7)}{\cos(\theta_6)} + \cos(\theta_3) - \cos(\theta_5)\cos(\theta_7) = 0$$

$$-\sin(\theta_3)\sin(\theta_4)-\sin(\theta_6)\sin(\theta_7)=0$$

$$\frac{5L\left(\sin(\theta_3)\cos(\theta_4)+\sin(\theta_7)\cos(\theta_6)-\cos(\theta_7)+2\right)}{2}=0$$

Set 2

$$-\frac{\sin^2(\theta_1)\sin(\theta_7)}{\cos(\theta_1)}-\sin(\theta_2)\sin(\theta_4)\sin(\theta_5)-\sin(\theta_3)\cos(\theta_2)\cos(\theta_5)-\sin(\theta_5)\cos(\theta_2)\cos(\theta_3)\cos(\theta_4)+\frac{\sin(\theta_7)}{\cos(\theta_1)}$$

$$\frac{\sin(\theta_1)\sin(\theta_6)\sin^2(\theta_7)}{\cos(\theta_7)}-\frac{\sin(\theta_1)\sin(\theta_6)}{\cos(\theta_7)}+\sin(\theta_2)\sin(\theta_4)\cos(\theta_3)-\cos(\theta_1)\cos(\theta_6)+\cos(\theta_2)\cos(\theta_4)=0$$

$$-\sin(\theta_1)\sin(\theta_7)-\sin(\theta_2)\sin(\theta_3)\cos(\theta_5)-\sin(\theta_2)\sin(\theta_5)\cos(\theta_3)\cos(\theta_4)+\sin(\theta_4)\sin(\theta_5)\cos(\theta_2)=0$$

$$\sin(\theta_3)\cos(\theta_4)\cos(\theta_5)+\sin(\theta_5)\cos(\theta_3)+\frac{\sin^2(\theta_6)\sin(\theta_7)}{\cos(\theta_6)}-\frac{\sin(\theta_7)}{\cos(\theta_6)}=0$$

$$-\sin(\theta_3)\sin(\theta_4)-\sin(\theta_6)\sin(\theta_7)=0$$

$$\sin(\theta_3)\sin(\theta_5)\cos(\theta_4)-\cos(\theta_3)\cos(\theta_5)+\cos(\theta_7)=0$$

$$\frac{5L\left(\sin(\theta_3)\cos(\theta_4)\cos(\theta_5)+\sin(\theta_3)\cos(\theta_4)+\sin(\theta_5)\cos(\theta_3)-\cos(\theta_7)+2\right)}{2}=0$$

Set 3

$$\frac{L\left(-4\sin(\theta_1)\sin(\theta_2)-5\sqrt{2}\sin(\theta_3)\cos\left(\theta_5+\frac{\pi}{4}\right)-5\sqrt{2}\sin\left(\theta_5+\frac{\pi}{4}\right)\cos(\theta_3)\cos(\theta_4)+4\cos(\theta_1)\cos(\theta_2)+2\right.}{2}$$

$$-\sin(\theta_3)\sin(\theta_4)\sin(\theta_6)+\sin(\theta_3)\cos(\theta_4)\cos(\theta_5)\cos(\theta_6)+\sin(\theta_5)\cos(\theta_3)\cos(\theta_6)-\sin(\theta_7)=0$$

$$\sin(\theta_3)\sin(\theta_5)\cos(\theta_4)-\cos(\theta_3)\cos(\theta_5)+\cos(\theta_7)=0$$

$$-\sin(\theta_3)\sin(\theta_4)\cos(\theta_6)-\sin(\theta_3)\sin(\theta_6)\cos(\theta_4)\cos(\theta_5)-\sin(\theta_5)\sin(\theta_6)\cos(\theta_3)=0$$

$$\frac{5L\left(-\sqrt{2}\sin(\theta_3)\sin\left(\theta_5+\frac{\pi}{4}\right)\cos(\theta_4)-\sin(\theta_3)\cos(\theta_4)+\sqrt{2}\cos(\theta_3)\cos\left(\theta_5+\frac{\pi}{4}\right)-2\right)}{2}=0$$

$$-\frac{\sin^2(\theta_1)\sin(\theta_2)\sin(\theta_7)}{\cos(\theta_1)}+\sin(\theta_1)\sin(\theta_7)\cos(\theta_2)+\frac{\sin(\theta_2)\sin(\theta_7)}{\cos(\theta_1)}-\sin(\theta_4)\sin(\theta_5)=0$$

$$-\sin(\theta_1)\sin(\theta_2)+\sin(\theta_4)\sin(\theta_6)\cos(\theta_5)+\cos(\theta_1)\cos(\theta_2)-\cos(\theta_4)\cos(\theta_6)=0$$

$$\frac{L\left(-4\sin(\theta_1)\cos(\theta_2)-4\sin(\theta_2)\cos(\theta_1)-2\sin(\theta_2)+5\sqrt{2}\sin(\theta_4)\sin\left(\theta_5+\frac{\pi}{4}\right)+5\sin(\theta_4)\right)}{2}=0$$

Set 4

$$-\sin(\theta_3)-\sin(\theta_4)\sin(\theta_6)\sin(\theta_7)-\sin(\theta_5)\cos(\theta_4)\cos(\theta_7)+\sin(\theta_7)\cos(\theta_4)\cos(\theta_5)\cos(\theta_6)=0$$

$$\sin(\theta_1)\sin(\theta_2)-\sin(\theta_4)\sin(\theta_6)\cos(\theta_5)-\cos(\theta_1)\cos(\theta_2)+\cos(\theta_4)\cos(\theta_6)=0$$

$$-\sin(\theta_4)\sin(\theta_5)\cos(\theta_7)+\sin(\theta_4)\sin(\theta_7)\cos(\theta_5)\cos(\theta_6)+\sin(\theta_6)\sin(\theta_7)\cos(\theta_4)=0$$

$$-\frac{\sin(\theta_1)\sin^2(\theta_2)\sin(\theta_3)}{\cos(\theta_2)}+\frac{\sin(\theta_1)\sin(\theta_3)}{\cos(\theta_2)}+\sin(\theta_2)\sin(\theta_3)\cos(\theta_1)+\sin(\theta_5)\sin(\theta_6)=0$$

$$-\sin(\theta_5)\sin(\theta_7)\cos(\theta_6)+\cos(\theta_3)-\cos(\theta_5)\cos(\theta_7)=0$$

$$\frac{L\left(-2\sin(\theta_3)\cos(\theta_2)-4\sin(\theta_3)+4\sin(\theta_5)\cos(\theta_6)\cos(\theta_7)-4\sin(\theta_7)\cos(\theta_5)-10\cos(\theta_3)+5\sqrt{2}\cos\left(\theta_5\right.\right.}{2}$$

Set 5

$$\frac{\sin(\theta_3)\sin^2(\theta_4)}{\cos(\theta_4)} - \frac{\sin(\theta_3)}{\cos(\theta_4)} - \sin(\theta_5)\cos(\theta_7) + \sin(\theta_7)\cos(\theta_5)\cos(\theta_6) = 0$$

$$-\sin(\theta_1)\sin(\theta_5)\sin(\theta_6)+\frac{\sin^2(\theta_2)\sin(\theta_3)}{\cos(\theta_2)}-\frac{\sin(\theta_3)}{\cos(\theta_2)}-\sin(\theta_5)\cos(\theta_1)\cos(\theta_6)\cos(\theta_7)+\sin(\theta_7)\cos(\theta_1)\cos(\theta_5$$

$$\sin(\theta_1)\sin(\theta_5)\cos(\theta_6)\cos(\theta_7)-\sin(\theta_1)\sin(\theta_7)\cos(\theta_5)-\sin(\theta_2)\sin(\theta_3)-\sin(\theta_5)\sin(\theta_6)\cos(\theta_1) = 0$$

$$\sin(\theta_5)\sin(\theta_7)\cos(\theta_6) - \cos(\theta_3) + \cos(\theta_5)\cos(\theta_7) = 0$$

$$\sin(\theta_1)\sin(\theta_6)\cos(\theta_7)+\frac{\sin(\theta_2)\sin^2(\theta_3)\sin(\theta_4)}{\cos(\theta_3)}-\frac{\sin(\theta_2)\sin(\theta_4)}{\cos(\theta_3)}+\cos(\theta_1)\cos(\theta_6)-\cos(\theta_2)\cos(\theta_4) = 0$$

$$\sin(\theta_3)\sin(\theta_4) + \sin(\theta_6)\sin(\theta_7) = 0$$

$$L\left(\sin(\theta_1)\cos(\theta_6) - 5\sin(\theta_3)\sin(\theta_4) + 2\sin(\theta_4)\cos(\theta_3) - \sin(\theta_6)\cos(\theta_1)\cos(\theta_7) - 2\sin(\theta_6)\cos(\theta_7)\right) = 0$$

Set 6

$$-\sin(\theta_3)\sin(\theta_4) - \sin(\theta_6)\sin(\theta_7) = 0$$

$$-\sin(\theta_1)\sin(\theta_2)\cos(\theta_6)+\sin(\theta_1)\sin(\theta_6)\cos(\theta_2)\cos(\theta_7)+\sin(\theta_2)\sin(\theta_6)\cos(\theta_1)\cos(\theta_7)+\cos(\theta_1)\cos(\theta_2)\cos$$

$$L\left(\sin(\theta_1)\cos(\theta_6) + 5\sin(\theta_6)\sin(\theta_7) - \sin(\theta_6)\cos(\theta_1)\cos(\theta_7) - 2\sin(\theta_6)\cos(\theta_7)\cos(\theta_1+\theta_2) - 2\sin(\theta_6)\cos\right.$$

$$-\frac{\sin(\theta_3)\sin^2(\theta_4)\sin(\theta_5)}{\cos(\theta_4)} + \frac{\sin(\theta_3)\sin(\theta_5)}{\cos(\theta_4)} - \cos(\theta_3)\cos(\theta_5) + \cos(\theta_7) = 0$$

$$\sin(\theta_1)\sin(\theta_7)\cos(\theta_2) + \sin(\theta_2)\sin(\theta_7)\cos(\theta_1) - \sin(\theta_4)\sin(\theta_5) = 0$$

$$\frac{L\left(5\sin(\theta_5) - 2\sin(\theta_7)\cos(\theta_1) - 4\sin(\theta_7)\cos(\theta_1+\theta_2) - 4\sin(\theta_7) - 10\cos(\theta_7) + 5\right)}{2} = 0$$

Set 7

$$\sin(\theta_1)\sin(\theta_2)\sin(\theta_3)\cos(\theta_7)-\sin(\theta_3)\cos(\theta_1)\cos(\theta_2)\cos(\theta_7)+\frac{\sin(\theta_5)\sin^2(\theta_6)}{\cos(\theta_6)}-\frac{\sin(\theta_5)}{\cos(\theta_6)}+\sin(\theta_7)\cos(\theta_3)$$

$$\frac{L\left(-10\sin(\theta_7) + 2\cos(\theta_1)\cos(\theta_7) - 5\cos(\theta_5)\cos(\theta_6) - 5\cos(\theta_6) + 4\cos(\theta_7)\cos(\theta_1+\theta_2) + 4\cos(\theta_7)\right)}{2} = 0$$

$$\sin(\theta_1)\sin(\theta_2)\sin(\theta_7)\cos(\theta_3)+\sin(\theta_3)\cos(\theta_7)-\frac{\sin^2(\theta_4)\sin(\theta_5)}{\cos(\theta_4)}+\frac{\sin(\theta_5)}{\cos(\theta_4)}-\sin(\theta_7)\cos(\theta_1)\cos(\theta_2)\cos(\theta_3)$$

$$-\sin(\theta_1)\sin(\theta_7)\cos(\theta_2) - \sin(\theta_2)\sin(\theta_7)\cos(\theta_1) + \sin(\theta_4)\sin(\theta_5) = 0$$

$$\sin(\theta_1)\sin(\theta_2)\sin(\theta_3)\sin(\theta_7) - \sin(\theta_3)\sin(\theta_7)\cos(\theta_1)\cos(\theta_2) - \cos(\theta_3)\cos(\theta_7) + \cos(\theta_5) = 0$$

$$\frac{L\left(-5\sin(\theta_5) + 2\sin(\theta_7)\cos(\theta_1) + 4\sin(\theta_7)\cos(\theta_1+\theta_2) + 4\sin(\theta_7) + 10\cos(\theta_7) - 5\right)}{2} = 0$$

$$\sin(\theta_1)\sin(\theta_2)+\frac{\sin(\theta_4)\sin^2(\theta_5)\sin(\theta_6)}{\cos(\theta_5)}-\frac{\sin(\theta_4)\sin(\theta_6)}{\cos(\theta_5)}-\cos(\theta_1)\cos(\theta_2)+\cos(\theta_4)\cos(\theta_6) = 0$$

$$\sin(\theta_1)\sin(\theta_3)\cos(\theta_2) + \sin(\theta_2)\sin(\theta_3)\cos(\theta_1) + \sin(\theta_5)\sin(\theta_6) = 0$$

$$\frac{L\left(-2\sin(\theta_1) + 5\sin(\theta_6)\cos(\theta_5) + 5\sin(\theta_6) - 4\sin(\theta_1+\theta_2)\right)}{2} = 0$$

5 Step 5: Single Driver

No analytical single driver found.

6 Step 6: Dual Driver

6.1 Input J3, J5

Found solution.

1. Solve θ_4 using Set 3, (p˙y)
   Form: $A\sin(\theta_4) + B\cos(\theta_4) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0$
   $A = 0, B = 0, C = \frac{5L\left(\sqrt{2}\cos(\theta_3)\cos\left(\theta_5+\frac{\pi}{4}\right)-2\right)}{2}$

2. Solve θ_7 using Set 2, (a˙z)
   Form: $A\sin(\theta_7) + B\cos(\theta_7) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0$
   $A = 0, B = 1, C = \sin(\theta_3)\sin(\theta_5)\cos(\theta_4) - \cos(\theta_3)\cos(\theta_5)$

3. Solve θ_6 using Set 1, (n˙z)
   Form: $A\sin(\theta_6) + B\cos(\theta_6) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0$
   $A = 0, B = 0, C = \tilde{\infty}\left(\frac{\sin^2(\theta_5)\sin(\theta_7)}{\cos(\theta_5)} - \frac{\sin(\theta_7)}{\cos(\theta_5)}\right) + \sin(\theta_3)\cos(\theta_4) + \sin(\theta_5)\cos(\theta_7)$

4. Solve θ_2 using Set 4, (p˙x)
   Form: $A\sin(\theta_2) + B\cos(\theta_2) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0$
   $A = 0, B = 0, C = \frac{L\left(10\sin(\theta_3)+4\sin(\theta_4)\sin(\theta_6)\cos(\theta_7)-4\sin(\theta_5)\sin(\theta_7)\cos(\theta_4)+5\sqrt{2}\sin\left(\theta_5+\frac{\pi}{4}\right)\cos(\theta_4)-4\cos(\theta_3)-4\cos(\theta_4)\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)\right.}{2}$

5. Solve θ_1 using Set 1, (n˙x)
   Form: $A\sin(\theta_1) + B\cos(\theta_1) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0$
   $A = \sin(\theta_2)\cos(\theta_3)\cos(\theta_4) - \sin(\theta_4)\cos(\theta_2), B = -\sin(\theta_2)\sin(\theta_4) - \cos(\theta_2)\cos(\theta_3)\cos(\theta_4), C = -\frac{\sin^2(\theta_5)\sin^2(\theta_6)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{\sin^2(\theta_5)\sin^2(\theta_6)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{\sin^2(\theta_5)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} - \frac{\sin^2(\theta_5)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \sin(\theta_5)\sin(\theta_7) + \frac{\sin^2(\theta_6)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} - \frac{\sin^2(\theta_6)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} - \frac{\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{1}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)}$

6.2 Input J4, J5

Found solution.

1. Solve θ_3 using Set 3, (p˙y)
   Form: $A\sin(\theta_3) + B\cos(\theta_3) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0$
   $A = 0, B = 0, C = -5L$

2. Solve θ_7 using Set 2, (a˙z)
   Form: $A\sin(\theta_7) + B\cos(\theta_7) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0$
   $A = 0, B = 1, C = \sin(\theta_3)\sin(\theta_5)\cos(\theta_4) - \cos(\theta_3)\cos(\theta_5)$

3. Solve θ_6 using Set 1, (n˙z)
   Form: $A\sin(\theta_6) + B\cos(\theta_6) + C = 0$
   Poly: $(C - B)t^2 + 2At + (B + C) = 0$
   $A = 0, B = 0, C = \tilde{\infty}\left(\frac{\sin^2(\theta_5)\sin(\theta_7)}{\cos(\theta_5)} - \frac{\sin(\theta_7)}{\cos(\theta_5)}\right) + \sin(\theta_3)\cos(\theta_4) + \sin(\theta_5)\cos(\theta_7)$

4. Solve θ_2 using Set 4, (p˙x)
   Form: $A\sin(\theta_2) + B\cos(\theta_2) + C = 0$
   Poly: $(C - B)t^2 + 2At + (B + C) = 0$
   $A = 0, B = 0, C = \frac{L\left(10\sin(\theta_3)+4\sin(\theta_4)\sin(\theta_6)\cos(\theta_7)-4\sin(\theta_5)\sin(\theta_7)\cos(\theta_4)+5\sqrt{2}\sin\left(\theta_5+\frac{\pi}{4}\right)\cos(\theta_4)-4\cos(\theta_3)-4\cos(\theta_4)\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)\right.}{2}$

5. Solve θ_1 using Set 1, (n˙x)
   Form: $A\sin(\theta_1) + B\cos(\theta_1) + C = 0$
   Poly: $(C - B)t^2 + 2At + (B + C) = 0$
   $A = \sin(\theta_2)\cos(\theta_3)\cos(\theta_4) - \sin(\theta_4)\cos(\theta_2), B = -\sin(\theta_2)\sin(\theta_4) - \cos(\theta_2)\cos(\theta_3)\cos(\theta_4), C = -\frac{\sin^2(\theta_5)\sin^2(\theta_6)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{\sin^2(\theta_5)\sin^2(\theta_6)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{\sin^2(\theta_5)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} - \frac{\sin^2(\theta_5)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \sin(\theta_5)\sin(\theta_7) + \frac{\sin^2(\theta_6)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} - \frac{\sin^2(\theta_6)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} - \frac{\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{1}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)}$

7 Step 7: Numerical Plots

No valid kinematic simulation generated.